

Winter Semester 2026/2027
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Exercise Session 1

Problem 1

1. Imagine that you are a manager of a public transportation company and you would like to find the optimal one-ride ticket price. To do so, you need to estimate the demand for travel by public transport as a function of the ticket price.
 - (i) Discuss the dependency of the demand of travel on fare price. What sign of the relationship do you expect? What shape of the relationship do you expect? What other factors (and why) might influence the demand for public transport?
 - (ii) Formulate a simple econometric model of the demand for public transport. What units of measurement would you use for each of the variables included? What is the economic/intuitive interpretation of the parameters in your model?
 - (iii) You managed to get the data on public transport in 10 cities:

City	Passengers per hour (thousand)	Fair price (dollars)
C_1	2073	0.85
C_2	2136	0.75
C_3	1879	0.60
C_4	938	1.00
C_5	7343	0.50
C_6	838	0.85
C_7	1648	1.00
C_8	739	0.75
C_9	1071	1.50
C_{10}	275	1.50

Using hand calculation or the program of your choice, estimate the demand for public transport. Assume the shape of the relationship as discussed in point (i), include just one explanatory variable.

- (iv) Using the data above and your estimates, calculate (or estimate) the following:
 - The value of the residual for each observation.
 - The sum of squared variation in the explanatory variable.
 - The variance of the disturbance term.

- The variance of the OLS estimator of the slope coefficient.
 - The t-statistics for the slope coefficient.
- (v) Using the data from the travel.csv let us estimate the model formulated in the previous part and discuss the properties of the OLS estimator.
- When the OLS estimator is unbiased? Which SLR/MLR assumptions should be imposed to prove the unbiasedness of the estimator?
 - When the OLS estimator is consistent? What is the difference between a consistent estimator and an unbiased estimator?
 - When the OLS estimator is efficient? Which MLR assumptions should be imposed to prove the efficiency of the estimator?
- (vi) Note that there is more information in the data than just ticket price and number of passengers. Discuss which of the additional variables might affect the number of passengers and why it might be useful to add them as control variables to the model. Estimate a multiple regression model and present results from different specifications in a form of a table.
- (vii) Interpret the results:
- What share of variation in the dependent variable did you explain?
 - What is the influence of each explanatory variable on the demand for public transport?
 - Are all the parameters of the same sign as you expected?
 - Which variables show to be significant in explaining the demand for public transport?

2. Goodness of fit:

- Show how R^2 is defined and provide the interpretation of this measure.
- What is the difference between R^2 and R_{adj}^2 ?

3. Discuss whether OLS assumptions are satisfied, use formal tests whenever possible:

- Check the normality of residuals. Use plots as well as a statistical test.
- Check the assumption of homoskedasticity. Use plots as well as a statistical test.